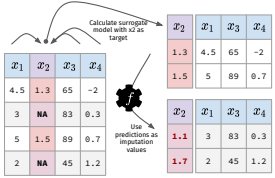
Applied Machine Learning

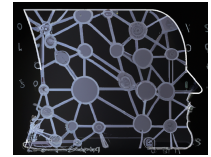
Imputation:
Advanced Methods and Pitfalls



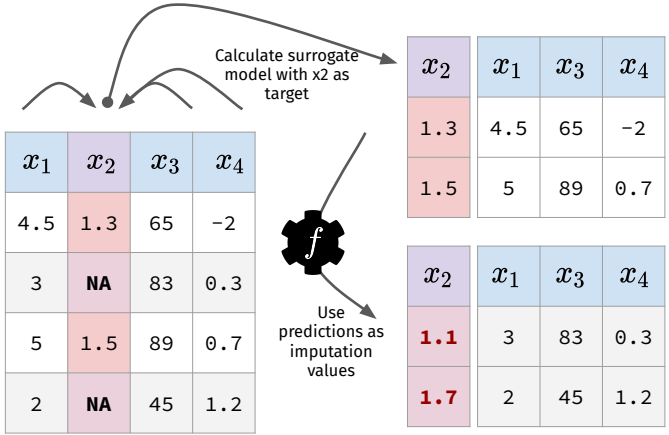
Learning goals

- Based imputation strategies
- Practical considerations for missing values

MODEL-BASED IMPUTATION



Instead of imputing a single value or sampling values it is desirable to take advantage of structure and correlation between features.

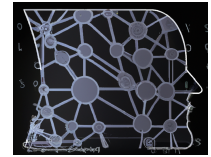


x_1	x_2	x_3	x_4
4.5	1.3	65	-2
3	**NA**	83	0.3
5	1.5	89	0.7
2	**NA**	45	1.2

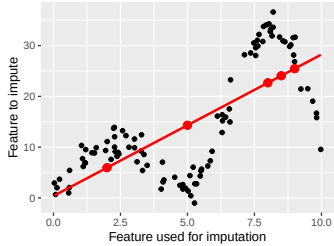
x_2	x_1	x_3	x_4
1.3	4.5	65	-2
1.5	5	89	0.7

x_2	x_1	x_3	x_4
1.1	3	83	0.3
1.7	2	45	1.2

MODEL-BASED IMPUTATION: DRAWBACKS

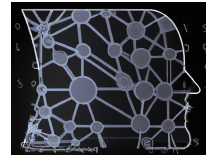


- Choice of surrogate model has high influence on the imputation:



- Surrogate model should handle missing values itself, otherwise imputation *loop* may be necessary.
- Surrogate model hyperparameters can be tuned and can be different for each feature to impute.

A TALE OF FAILED IMPUTATION



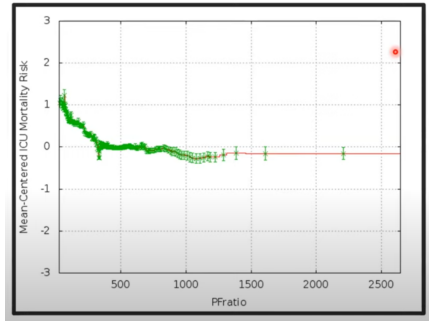
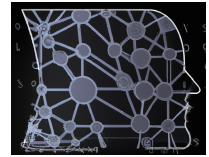
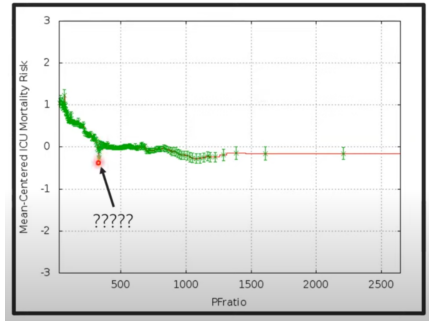


Figure: Figure from *Rich Caruana's talk at AutoML Conference 2024* example starting at 43'31

A TALE OF FAILED IMPUTATION 2





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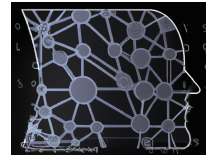
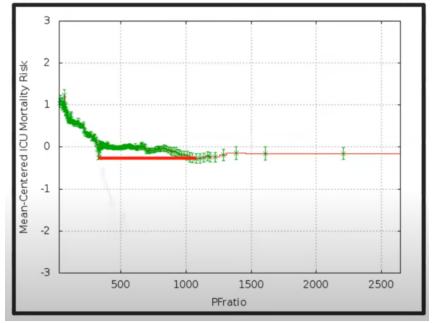
A TALE OF FAILED IMPUTATION 3





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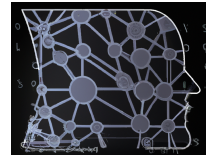
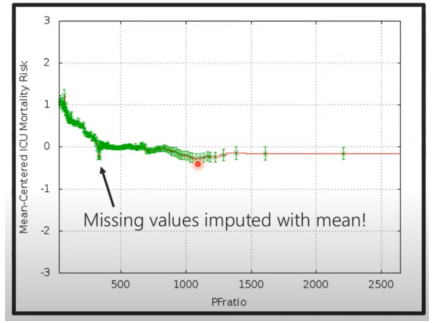
A TALE OF FAILED IMPUTATION 4





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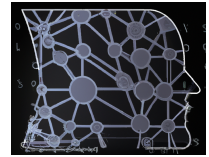
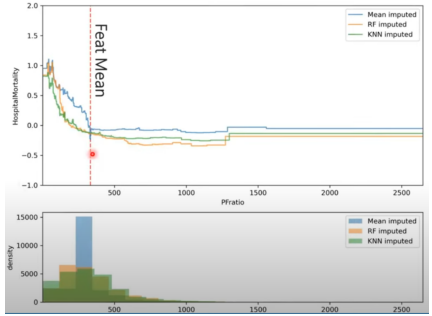
A TALE OF FAILED IMPUTATION 5





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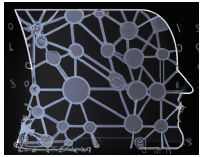
TESTS FOR MISSINGNESS MECHANISMS

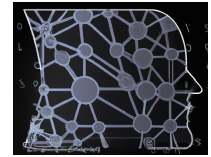


MAR vs MCAR

- Little's test
- Logistic regression for missingness
  1. For each column, create a new classification task: predict if the value is missing or not
  2. Use the remaining columns as features
  3. Fit logistic regression
  4. Examine coefficients and p-values

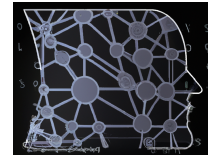
MNAR: no test available

CENSORING MECHANISM



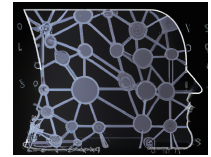
- **Left censoring:** True value below a threshold; exact value unknown.
- **Interval censoring:** True value lies within a known interval.
- **Right censoring:** True value above a threshold; exact value unknown.

CENSORING MECHANISM



- **Left censoring:** True value below a threshold; exact value unknown.
- **Interval censoring:** True value lies within a known interval.
- **Right censoring:** True value above a threshold; exact value unknown.
- **Type I censoring:** Experiment ends at pre-specified time; remaining subjects are right-censored.
- **Type II censoring:** Experiment ends after a fixed number of failures; remaining subjects are right-censored.
- **Random (non-informative) censoring:** Censoring times independent of failure times; observe minimum of failure and censoring time.

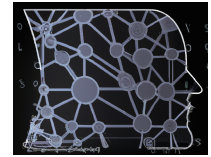
TAKEAWAYS OF A RECENT BENCHMARK



Imputation for prediction: beware of diminishing return. Le Morvan and Varoquaux, ICLR 2025 ▸ Morvan and Varoquaux 2024

- Better imputation performance does not result in better prediction performance for MNAR (w/o missingness indicator)
- Better imputation might be irrelevant b/c information is available in other features, unimportant, even with imputation, it might be hard to learn a good downstream model
- More expressive models benefit less from imputation
- Best on average: missForest + XGBoost + masking
- Open Questions:
  - performance of random draws
  - performance of multiple imputation
  - impact of performance on explainability and calibration
  - performance of models that can learn directly with missing value (advanced transformer architectures)

TAKEAWAYS



- There exist different reasons for missing data
- There exist different missingness mechanisms
- Different downstream tasks ask for different imputation strategies
- Different data modalities ask for different imputation strategies
- Understanding the data generating process helps deciding on the imputation strategy
- Common imputation strategies:
  - Constant value: mean, median
  - Imputation by sampling
  - Missingness indicator
  - Model-based

  $\rightarrow$ imputation selection is a CASH problem